

Amritraj Lamba

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EDUCATION

Heritage International Xperiential School, Gurugram.

Relevant Coursework: HL Math Analysis and Approaches, HL Chemistry, HL Physics, SL Economics, SL French B, SL English A Language and Literature

IGCSE (Grade 10, Academic Year 2023-24) : 10/10 A*, School Topper in Additional Mathematics, School Topper in Chemistry

SAT- 1550 (Maths 800, Reading & Writing 750)

AP- Calculus BC: 5/5

IELTS - Band 8

LEADERSHIP

Founder, Sewage Suraksha

April 24 - Present

- Developed a sewage lid with multi-gas sensors; it is a low-cost retrofit ensuring safety of municipal workers.
- The lid detects harmful gases, including H₂S, CH₄, NH₃, & CO, and alerts authorities to hazardous gas levels. It has been tested on 17 sewers and has been cleared by the National Disaster Management Authority for adoption.

Founder, Flood Shield

July 24 -Present

- Flood detection IOT for real-time cloud mapping and emergency coordination; Special Praise for "practicality" at the Princetonian. Adopted by the National Disaster Management Authority.
- Designed and piloted an affordable early warning system for floods that alerts authorities to areas where flooding occurs; adopted by 1000+ houses. Commended by the Haryana Government.
- Also received the Crest Gold Award for this invention.

Leader—Ingenix School STEM Society

Oct 24 - Oct 25

- Revived and expanded society from 20 to 50+ members.
- Mentored teams for the Cambridge Science Competition & F1 in Schools, helped them qualify for nationals.

Leader - "FUNCTION", After School Robotics Club

Oct 24 - Oct 25

- Founded a 20+ member club teaching CAD, programming, and mechanical design outside of competition teams.
- Introduced collaborative engineering workshops (bridge/tower builds, quizzes) to strengthen teamwork and experimentation.
- Built a hands-on learning culture that became an entry pathway for beginners; several members progressed to competitive robotics.

RESEARCH

Automation & Robot Adoption in Delhi-NCR Factories, Author

Jan 25- Sept 25

- Analysed five factories from fully manual (e.g., Metaval Engineering) to highly automated (e.g., Kadence Automation) to assess efficiency, training needs, and scalability.
- Found cobot-assisted systems improved efficiency by up to 20% while reducing fatigue and errors; highlighted the need for retraining, ergonomic design, and phased adoption.
- Developed a roadmap for SME modernisation without worker displacement; validated through Delhi-NCR field data, mentorship from Stanford professor Dr. Arvind Karunakaran, and publication in Lex Localis.

Coral Reefs as Natural Barriers to Coastal Erosion in India, Author

March 24- Aug 25

- Used satellite imagery and GIS to show shorelines protected by healthy reefs had less than half the erosion of areas with degraded reefs.
- Provided evidence to integrate reef restoration into climate adaptation strategies for cities like Mumbai and Chennai, complementing engineered defences.
- Recognition included publication in the International Journal of Environmental Sciences and mentorship from Professor Dr. Umang Soni, with validation in competitive research platforms.

SUMMER PROGRAMS

University of Pennsylvania, Wharton Global Youth Program

June 2025

Product Design Academy

- Collaborated with Professor Taylor Caputo and international peers under the Wharton faculty, learning design thinking, prototyping and TAM/SAM/SOM analysis.
- Learned to apply structured market-sizing frameworks to test creative engineering ideas against commercial viability. Prototypes developed during the program reflected this rigorous approach, and faculty feedback validated the professional grounding in product innovation.

Euler's Circle

June- July 2024

- "Serious Mathematics for Serious High School Students" under Stanford University Prof. Simon Salzedo.
- One of 30 members worldwide.
- Capstone on linear algebra and projective geometry; improved modeling intuition later applied to CFD and robot mechanisms.
- Attended on a 50% scholarship.

LaunchX, Online Bootcamp

June 2025

- Selected for LaunchX; collaborated with peers to ideate, prototype, and pitch ventures, gaining hands-on experience in market validation, business modelling, and product design.

INTERSHIPS

Summer Intern, Vardan Envirolabs, Delhi

Dec 24 - Feb 25

- Identified available and capable employees as part of the talent search module team, designed to assist talent recruitment in companies.
- Conducted market research for the firm, tapping into new domains such as the legal sector, to streamline services and the employment data collection process.

ACADEMIC HONORS & AWARDS

MATHEMATICS

USA Math Olympiads (2022-24)

- AMC 12B (2024): Scored 132/150
- School Rank 1, Country Rank 28, Qualified for AIME, Certificate of Distinction, Honour Roll (Top 2.5%)

World International Mathematics Olympiad (WIMO), Final, January 2024 (Kuala Lumpur, Malaysia)

- Bronze Award

CEMC (University of Waterloo)

- Pascal Contest (9th Grade, 2023): 2nd Place from India region, Certificate of Distinction, Student Honour Roll
- Fryer Contest (9th Grade, 2023): 2nd Place from India region, Certificate of Distinction
- Cayley Contest (10th Grade, 2024): Certificate of Distinction, Student Honour Roll
- Galois Contest (10th Grade, 2024): Certificate of Distinction

- Fermat Contest (11th Grade, 2025): 9th Place from India region, Certificate of Distinction, Student Honour Roll
- Hypatia Contest (11th Grade, 2025): Certificate of Distinction

SIMOC (Singapore International Mathematical Olympiad Challenge), Singapore, July 2024

- Maths Olympiad Contest: Silver Award
- Maths Warrior Challenge: Silver Award
- Mind Sports Challenge: Bronze Award

IOQM (Indian Olympiad Qualifier in Mathematics) 2024–25

- Certificate of Merit

SASMO (Singapore and Asian School Mathematics Olympiad) 2024 (Grade 10)

- Gold Award

AMO (American Mathematics Olympiad)

- Gold Award (2023, Grade 10)
- Gold Award (2022, Grade 9)

TIMO (Thailand International Mathematical Olympiad) Final Round 2022–23

- Gold Award (Secondary 3 Group)

HKIMO (Hong Kong International Mathematical Olympiad)

- Heat Round 2024: Gold Award (Senior Secondary Group)
- Final Round 2022: Gold Award (Secondary 2 Group)

Stanford Math Tournament (SMT) 2023

- Honourable Mention in General Test
- Australian Mathematics Competition 202
- Certificate of Distinction (Senior Division, Grade 11)

International Maths Kangaroo Competition (2023)

- Silver Medal

International Junior Honor Society (IJHS), Singapore (2023/2024)

- Certificate of Membership

CHEMISTRY

Avogadro Chemistry Contest (University of Waterloo) (11th Grade, 2025)

- Certificate of Distinction

PHYSICS

Sir Isaac Newton Physics Contest (University of Waterloo) (11th Grade, 2025)

- Certificate of Distinction

Future Intelligence Science Olympiad Physics, International Round (2024)

- Gold Medal

ROBOTICS

First Tech Challenge (FTC), February–July 2024

- Role: CAD and Design Head of Team, Game Player
- Achievements: Qualified for Asian Pacific Open Championship (APOC), Sydney, Australia
- Won: Inspire Award (2nd Place) and Think Award 2024, 4th Ranked in Knockout Races out of 32 teams.

STEM

F1 in Schools 2022- 2024

- Role: Design Engineer
- Ranked 15th in Nationals
- Regional Digital Marketing Award

VOLUNTEER WORK

Teens of God (Regd. Local NGO) (Aug 2022 - Dec 2023)

- Mentor/Tutor teaching basic Math and Science concepts to underprivileged children.
- One of the lead members of the NGO's robotics branch, where we taught concepts of robotics and circuits using donated STEM kits.

Signal Stars (FTC Outreach Initiative, Thane) (Jan 2024 – Jul 2024)

- Volunteered as a mentor for the Signal Stars team from a rural-area community in Thane.
- Led rural students through design, construction and competition preparation, culminating in the team earning the Judges' Choice Award at the FIRST Tech Challenge India Championship 2023-24.
- Guided the students to present a strong engineering portfolio, build a modular robot mechanism and collaborate effectively under competition constraints.

SKILLS

- **Technical Skills/Tools:** Python, 3D Modelling (CAD), Arduino, Robotics
- **Sports:** Diploma & Black Belt in Shotokan Karate.
- **Music:** Guitar (Trinity Rock & Pop Grade 5)
- **Languages:** English, French, Hindi, Punjabi